2> $\|u\| = \sqrt{u \cdot u}$

$= \sqrt{\langle u \cdot u \rangle}$

$\|u\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$

$= \sqrt{x_1 x_1 + x_2 x_2 + \cdots + x_n x_n}$

$= \sqrt{(x_1, x_2, \cdots, x_n)(x_1, x_2, \cdots, x_n)}$

$= \sqrt{u \cdot u}$

$= \sqrt{\langle u, u \rangle}$

$\|\hat{u}\| = \left\| \frac{u}{\|u\|} \right\| = \frac{\|u\|}{\|u\|} = 1$

$\|f\| = \sqrt{\langle f, f \rangle}$

$= \left(\int_a^b f^2(t)\, dt \right)^{1/2}$

$\hat{f} = \frac{f}{\|f\|}$

18.02.2020

* $u \cdot v = \|u\| \|v\| \cos\theta$

$\cos\theta = \frac{u \cdot v}{\|u\| \|v\|}$

$\Rightarrow \theta = \cos^{-1}\left(\frac{u \cdot v}{\|u\| \|v\|} \right)$

If $\frac{u \cdot v}{\|u\| \|v\|} = 0 \Rightarrow u \cdot v = 0$

$\Rightarrow \langle u, v \rangle = 0$

Orthogonal vectors

Let V be an inner product space. The vectors $u, v \in V$ are said to be orthogonal if $\langle u, v \rangle = 0$.

Orthonormal vectors

Let V be an inner product space. The vectors $u, v \in V$ are said to be orthonormal if (i) $\langle u, v \rangle = 0$ (ii) $\|u\| = \|v\| = 1$

Pkalika Maths